Johan Ajnabi

PhD Candidate | Wound Healing, Mechanobiology and Epigenetics

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem)
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Summary

Mechanobiology and epigenetics researcher investigating how mechanical forces regulate chromatin organization and cell fate transitions during tissue repair. Identified a mechanosensitive regulatory mechanism controlling DNMT3a nucleocytoplasmic partitioning and linking cytoskeletal dynamics to epigenetic reprogramming. Integrates quantitative imaging, molecular perturbation, and cell biology to define how tissue architecture governs regeneration and disease.

Education

PhD Candidate, BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem) — *2019 - Present*
Advisor: Prof. Colin Jamora
PhD thesis: Regulation of DNMT3a localization in the cutaneous wound healing response

M.Sc. in **Molecular Biology and Biotechnology**, ICAR-Indian Agricultural Research Institute — *2019*

B.Sc. (Hons.) in **Agriculture**, Bidhan Chandra Krishi Viswavidyalaya — *2017*

Research Experience

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem), India — *2019-Present*

Research Scholar (PhD)
Advisor: Prof. Colin Jamora

Mechanotransduction and epigenetic regulation (Primary project)

- Identified a mechanosensitive regulation mechanism of DNMT3a nucleocytoplasmic shuttling, establishing a link between cytoskeletal dynamics and epigenetic reprogramming during wound repair
- Integrated live-cell imaging, biochemical assays, and genetic perturbations to define ERK-dependent mechanotransduction pathways
- Led first-author study (bioRxiv: Ajnabi *et al.,* 2026).

Stem cell signaling and niche regulation

- Contributed to identifying a Mindin-integrin-STAT3 signaling axis that sustains keratinocyte stemness through integrin endocytosis (Cell Communication and Signaling: Dam *et al.,* 2026).
- Contributed to work showing how autocrine Mindin signaling maintains stem and progenitor states in skin epithelium and carcinoma cells (Cell Reports: Badarinath *et al.,* 2022).

Tissue-scale and translational biology (vitiligo, fibrosis)

- Performed spatial transcriptomic profiling of vitiligo skin before and after NB-UVB treatment and validated immune-state changes by immunofluorescence and qPCR (Clinical & Translational Immunology: Dutta *et al.,* 2026).
- Studied extracellular matrix regulation of cutaneous fibrosis using transgenic mouse models (Journal of Investigative Dermatology: Rana *et al.,* 2023).

Host-pathogen interactions

- Investigated LL-37-mediated inhibition of SARS-CoV-2 entry using mammalian cell systems, pseudovirus assays, and flow cytometry (Frontiers in Immunology: Bhatt *et al.,* 2023).

ICAR-National Institute for Plant Biotechnology, India — *2017-2019*

M.Sc. dissertation
Advisor: Dr. Monika Dalal

- Characterized promoter cis-regulatory regions controlling *PM19* expression in wheat using molecular cloning, transient GUS assays, plant tissue culture, transformation, and sequence analysis (Thesis: Ajnabi J, 2019)

Publications

First-Author & Lead Contributions

- **Ajnabi J**, Dam B, Gupta E, Saha T, et al. *Actin-dependent mechanotransduction controls nucleocytoplasmic partitioning of DNMT3a through ERK1/2 signaling during cutaneous wound healing.* **bioRxiv**, 2026 *(preprint)*

Collaborative Publications

- Dam B**, Ajnabi J**, et al. *Mindin-mediated αM-integrin endocytosis activates STAT3 to maintain keratinocyte stemness.* **Cell Communication and Signaling**, 2026
- Dutta A, Gupta D, **Ajnabi J**, et al. *Spatial transcriptomic analysis of the immune landscape following NB-UVB treatment of vitiligo skin.* **Clinical & Translational Immunology**, 2026
- Dutta S, Islam Z, Das S, Barman A, Chowdhury M, Mondal BP, **Ajnabi J**, et al. *Harmonizing plant resilience: membrane lipid dynamics in response to abiotic stresses.* **Discover Plants**, 2025
- Bhatt T, Dam B, Khedkar SU, Lall S, Pandey S, Kataria S, **Ajnabi J**, et al. *Niacinamide enhances cathelicidin-mediated SARS-CoV-2 membrane disruption.* **Frontiers in Immunology**, 2023
- Rana I, Kataria S, Tan TL, Hajam EY, Kashyap DK, Saha D, **Ajnabi J,** et al. *Mindin (SPON2) is essential for cutaneous fibrogenesis in a mouse model of systemic sclerosis.* **Journal of Investigative Dermatology**, 2023
- Badarinath K, Dam B, Kataria S, Zirmire RK, Dey R, Kansagara G, **Ajnabi J**, et al. *Snail maintains the stem/progenitor state of skin epithelial cells and carcinomas through autocrine Mindin signaling.* **Cell Reports**, 2022

Fellowships and Awards

- **Paeonia Foundation Travel Award –** Selected to present research at *Mechanobiology in Health and Disease Conference*, National University of Singapore, 2023
- **ASRB-NET (Agricultural Biotechnology)** – Qualified national eligibility test for lectureship, 2020
- **CSIR-NET Junior Research Fellowship (JRF) (AIR 20)** – National-level fellowship awarded to top-ranked candidates by the Council of Scientific and Industrial Research (CSIR), India, 2019
- **ICMR Junior Research Fellowship (JRF)** – Competitive national fellowship in Life Sciences by the Indian Council of Medical Research (ICMR), 2019
- **GATE XL (AIR 8)** – All India Rank 8 in Graduate Aptitude Test in Engineering (Life Sciences), 2019
- **DBT-BCIL Fellowship** – National-level fellowship awarded by the Department of Biotechnology, Government of India, 2019
- **ICAR AICE-JRF/SRF (AIR 3) –** All India Rank 3 in national examination for agricultural biotechnology research fellowships by the Indian Council of Agricultural Research (ICAR), 2019
- **ICAR AIEEA-PG (AIR 1)** – All India Rank 1 in national entrance examination for postgraduate studies in agricultural biotechnology, 2017

Selected Presentations

- **Poster** – "Epigenetic and mechanical regulation of the cutaneous wound healing response" at *Mechanobiology in Health and Disease Conference*, National University of Singapore, Singapore, 2023
- **Poster** – "Role of DNMT3a in the cutaneous wound healing response using a mouse model" at *10th International Conference of Laboratory Animal Scientists' Association (LASA)*, Hyderabad, India, 2022
- **Poster** – "Identification and characterization of stress-responsive PM19 promoter from wheat" at *National Agricultural Science Congress,* New Delhi, India, 2019

Technical Expertise

Experimental Biology

- Experimental design and hypothesis-driven research
- Primary epidermal keratinocyte culture, mouse models, and colony management
- Quantitative live-cell and fluorescence microscopy (confocal and multiphoton), tissue imaging, and flow cytometry (FACS)

- Molecular and cellular biology: genetic perturbation, transfection, qPCR, Western blotting, pseudovirus generation and assays, BSL-2 workflows

Computational & Analytical
- GraphPad Prism, Adobe Illustrator, MATLAB, R (basic), and Python (basic)
- Image analysis (FIJI/ImageJ)

Mentoring & Collaboration

- Mentored junior researchers in primary cell culture, molecular biology techniques, and experimental design, including training in data interpretation and scientific communication (presentations and writing)
- Collaborated on interdisciplinary projects spanning mechanobiology, immunology, and spatial transcriptomics, integrating experimental and clinical datasets
- Contributed to multi-author studies across epithelial biology, fibrosis, and host–pathogen interactions, supporting experimental design, data generation, and analysis

Languages

- Bengali (Native)
- English (Advanced)
- Hindi (Limited working proficiency)

References

Prof. Colin Jamora
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Department of Life Sciences, Shiv Nadar Institution of Eminence Deemed to be University, Gautam Buddha Nagar, Tehsil Dadri, Uttar Pradesh - 201314, India
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